

Computer vision

Introduction

Vanda Benešová

Let me introduce myself...

Prof. Ing. Vanda Benešová CSc.

Montanuni Leoben -CD Labor

Muhlegger G.m.b.H. Graz and LEM Norma

Joanneum Research Graz 2000-2010

STU FIIT Bratislava until 2011

lectures:

- Human Computer Interaction
- Computer vision
- Digital image processing, multimedia and graphics
- AI v HCI (FEL ČVUT)



Rules

max 24p. **exercises**

max 16p. **test**

!!! Minimum **20p.** (from max 24p+16p.) to get preliminary pass (zápočet)

max 60p. **final exam**

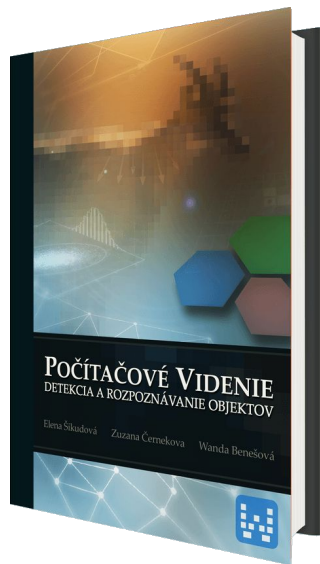
!!! Minimum **30p.** (from max 60p.) to pass the exam

Exercises

1st assignment:	1. - 3. exercise	max 6p
2nd assignment:	4. - 6. exercise	max 6p
3rd assignment:	7. - 9. exercise	max 6p
4th assignment:	10.- 12. exercise	max 6p

		max 24 p

Recommended readings



Počítačové Videnie

Detekcia a rozpoznávanie objektov

E. Šikudová, Z. Černeková, W. Benešová, Z.
Haladová, and J. Kučerová
Praha: Wikina Praha, 2013, p. 397

Recommended readings

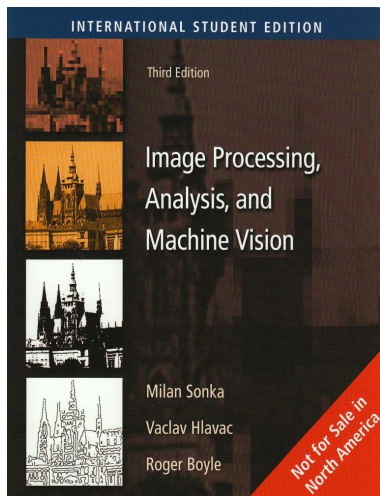
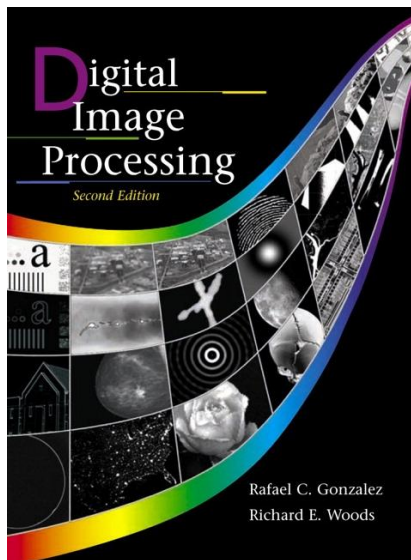


Image Processing, Analysis and Machine Vision

Milan Šonka, Václav Hlaváč,
Roger Boyle

Recommended readings



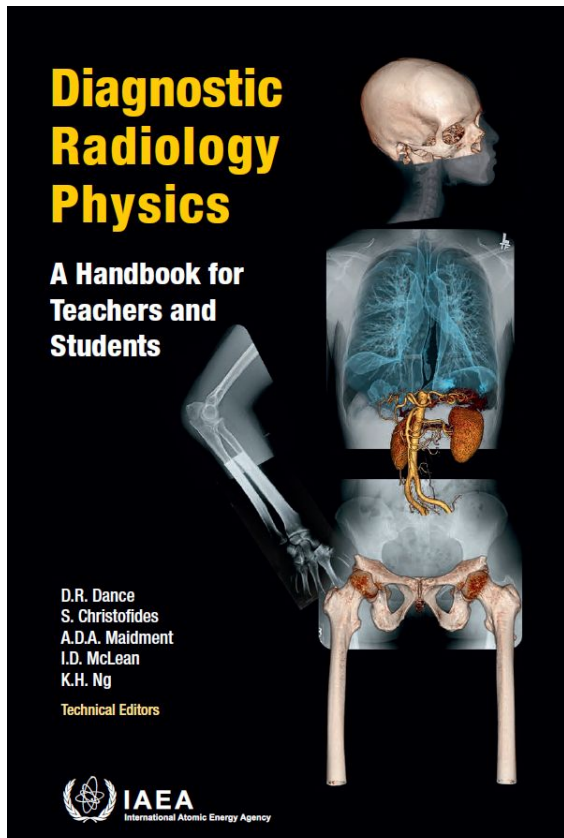
Digital Image Processing

Rafael C. Gonzalez, University of Tennessee
Richard E. Woods, MedData Interactive

Recommended readings

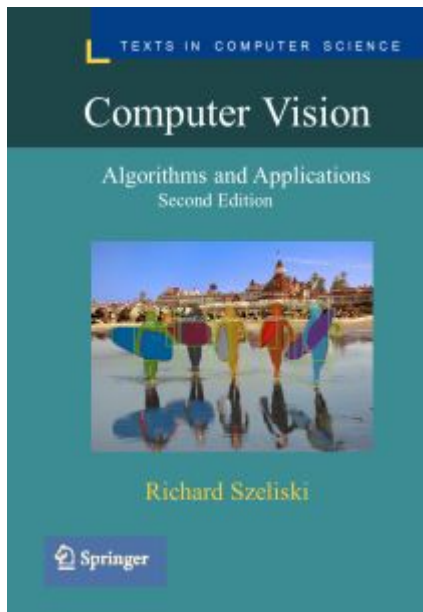
Diagnostic Radiology Physics: A Handbook for Teachers and Students

D.R. Dance
S. Christofides
A.D.A. Maidment
I.D. McLean
K.H. Ng



Recommended readings

<https://szeliski.org/Book/>



PIV lectures 2025

Intro, Image Enhancement

Morphological processing

Image Filtration

Color

DNN (Deep Neural Network) in Computer vision, Interpretability

Object detection and recognition

Local Descriptors

Segmentation

Motion detection

3D reconstruction - intro

Image Registration

Medical imaging

Saliency + recap.

Scope of the introduction (1.lecture)

- What is Computer vision?
- Application examples
- Visual data acquisition

What is Computer vision?

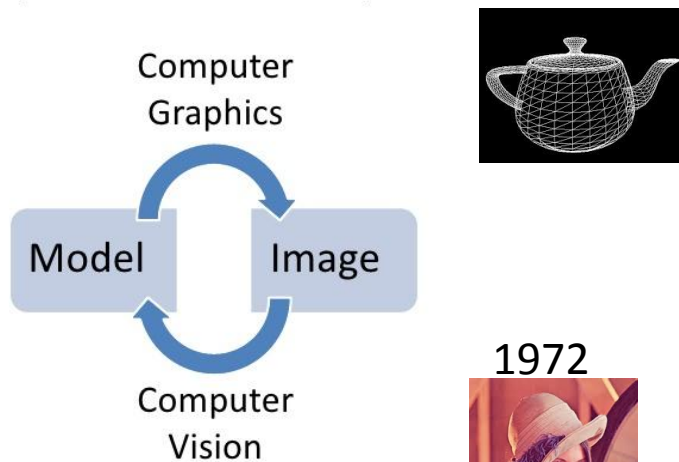
What is „Computer vision“?

Computer vision is concerned with the automatic extraction, analysis and understanding of useful information from a single image or a sequence of images or other visual data.

It involves the development of a theoretical and algorithmic basis to achieve automatic visual understanding.

What is „Computer vision“?

Synthesis vs Analysis



1972



1997



Computer graphics vs. Computer vision

Computer vision, Machine vision, Image processing & Related topics

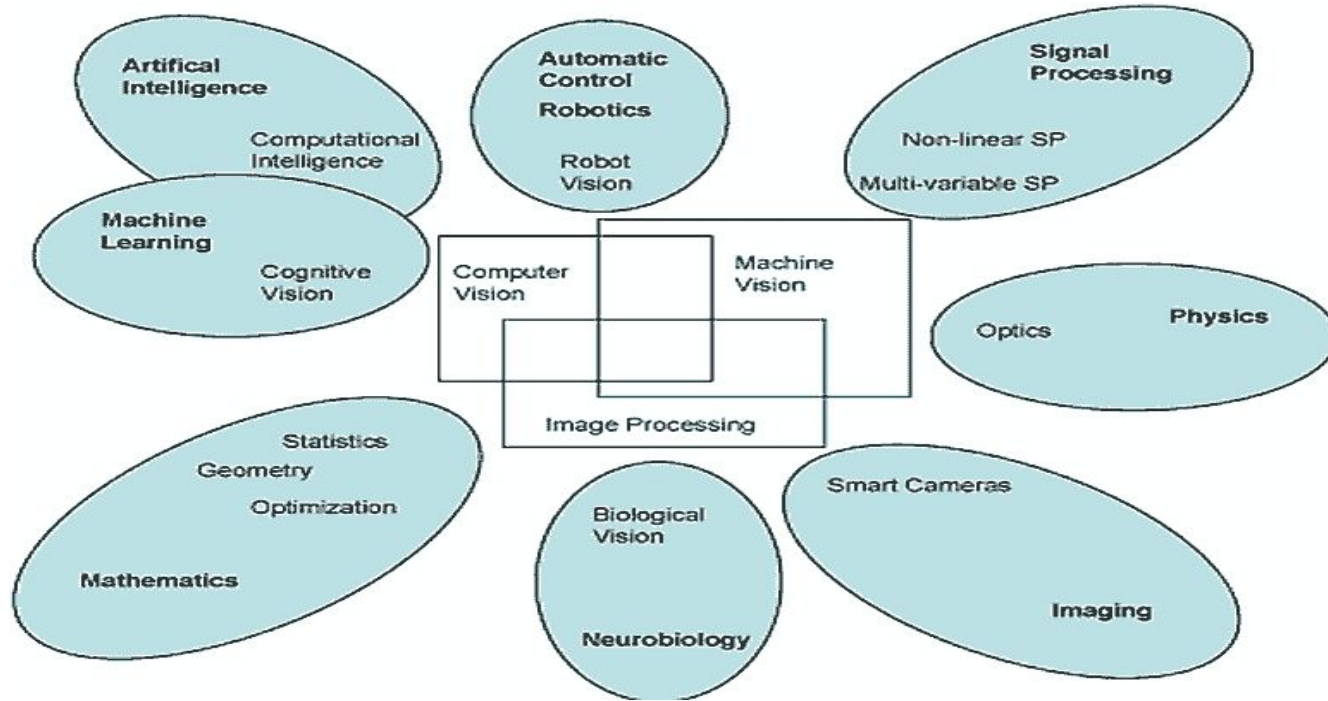


Image processing

Low level operations:

- Image preprocessing
- Noise reduction
- Contrast enhancement
- Edge detection
- Morphological image processing
- Image sharpening
- Colour image processing
- Transforms (Fourier, Cosine, Wavelet)
- Image compression

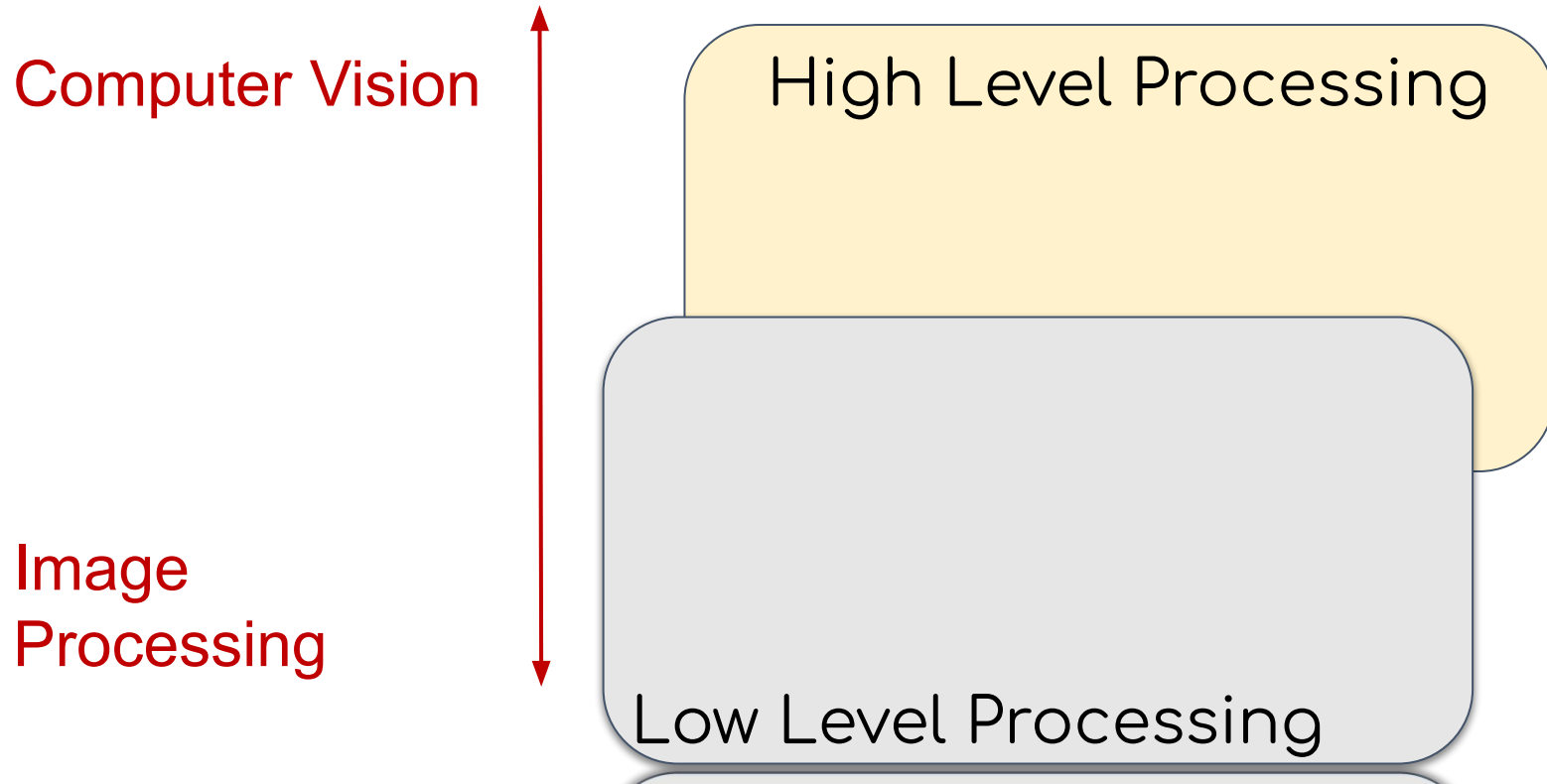
A low level process is characterized by the fact that both its inputs and outputs are images

Computer vision

High level processing -> extraction of information and more...

- Object detection
- Object recognition, Pose recognition
- Segmentation
- Image registration
- 3D object reconstruction
- Object tracking
- Motion detection

Image processing -> Computer vision



Application examples

Applications fields of Computer Vision

- image restoration
- augmented reality AR
- autonomous vehicles
- character recognition OCR
- biometrics:
 - fingerprint identification
 - face recognition
 - gesture analysis
- human-computer interaction HCI
- medical imaging
- remote sensing, geoscience
- security, forensics
- industrial quality inspection
 - process control
- robotics

Industry applications examples

Industrial quality inspection

Process control

- Quality inspection
- Shape (2D/3D measurement)
- Position
- Surface
- OCR
- Measurement

Industrial quality inspection - examples

3D vision system

Defects in specular surfaces



Phase measurement technique detects defects in specular surfaces

The human eye is certainly capable of identifying even the smallest topographical changes on the surfaces of objects; however, the nature of their surfaces often requires a particular automated imaging system for detailed analysis. A number of different methods have been used to measure the surface of diffusely reflecting objects, including structured-light projection, fringe pattern projection and stereo vision techniques.

However, when the surfaces of objects such as automotive body panels and optical components are specularly reflective, such techniques will not prove as effective. While alternative techniques such as white light interferometry can be used to assess surface defects, they are often limited by their small field of view.

Now, Stefan Werling and his colleagues at the Fraunhofer Institute of Optronics (www.iosb.fraunhofer.de) have developed a technique known as phase measurement deflectometry (PMD) to overcome these limitations.

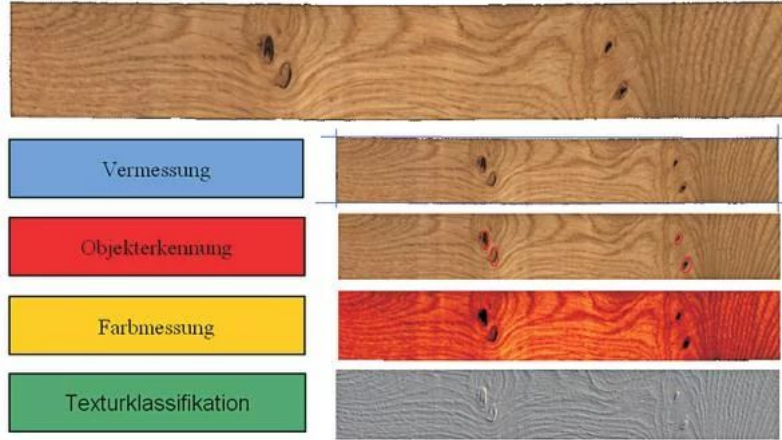
Specular objects reflect light in a single direction, so incoming light must be provided from many directions to ensure that some light is reflected into the camera system.

Because of *Phase measurement continued on page 10*



FIGURE 1. At Fraunhofer IOSB, Stefan Werling and his colleagues have developed a system based on phase measurement deflectometry (PMD) to measure specularly reflective parts such as painted automotive panels.

Industrial quality inspection - examples



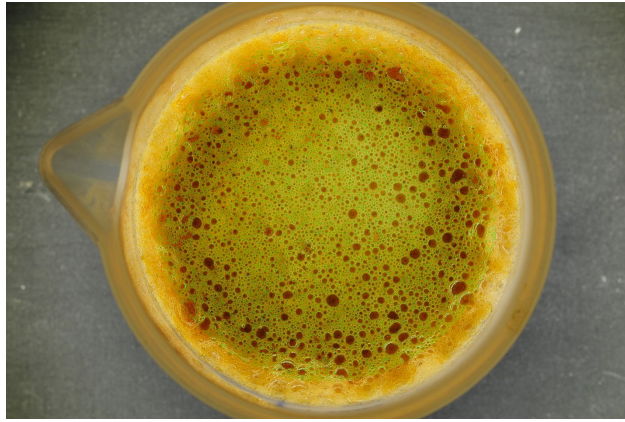
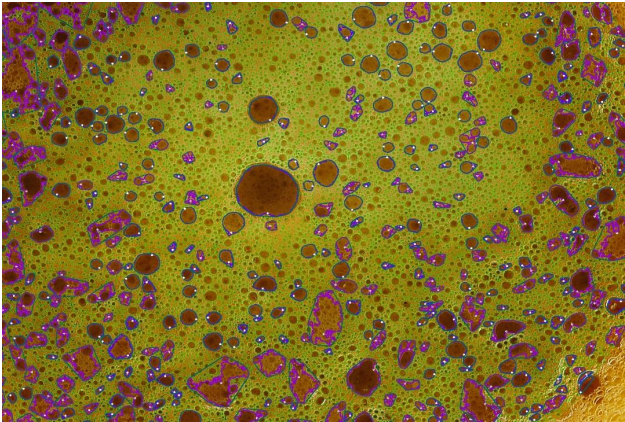
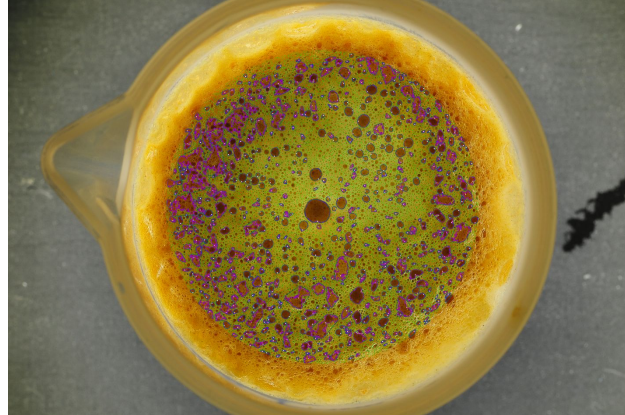
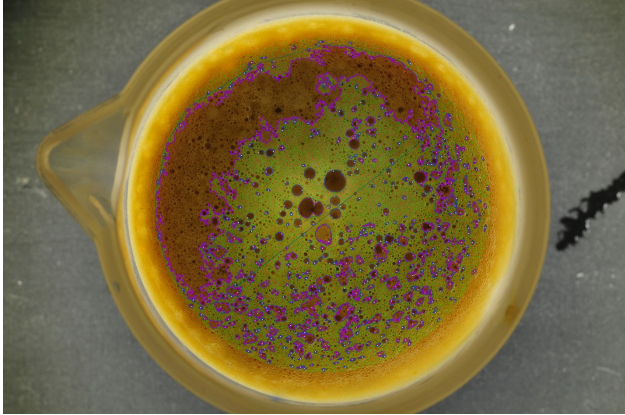
<https://www.joanneum.at/digital/>

Machine vision system aids in high-speed beer bottle inspection process

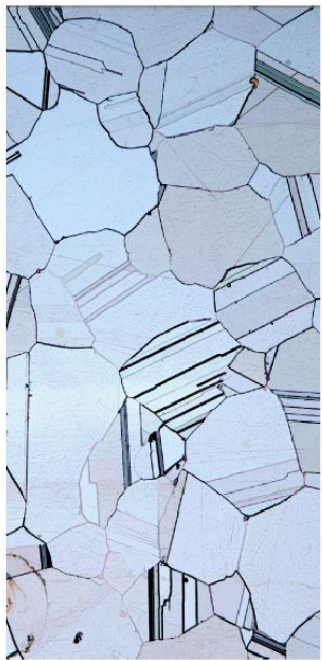
22 bottles/sec



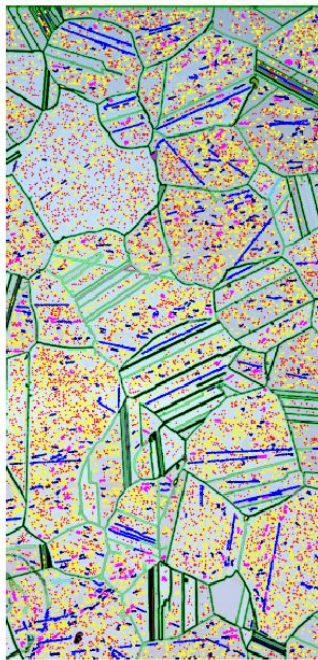
Industrial quality inspection - examples



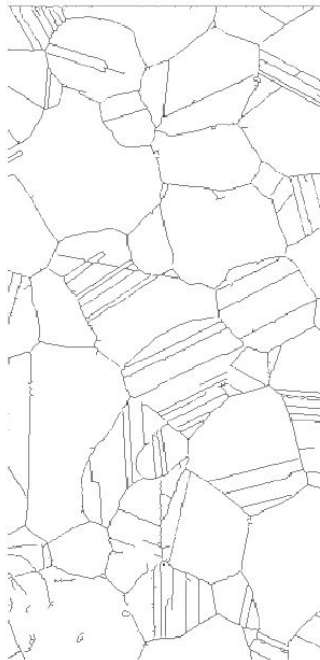
Micrographs of super-alloy: segmentation



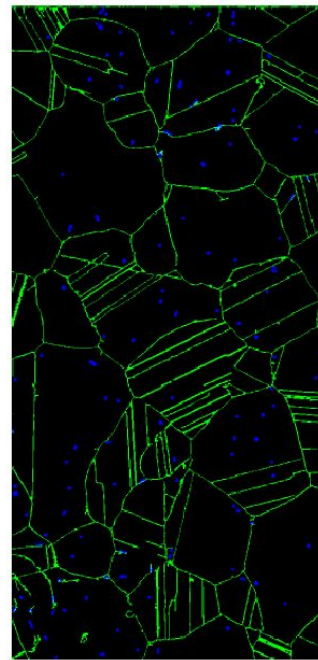
a) Vstupný obraz -
príklad



b) Výsledok klasi-
fikácie

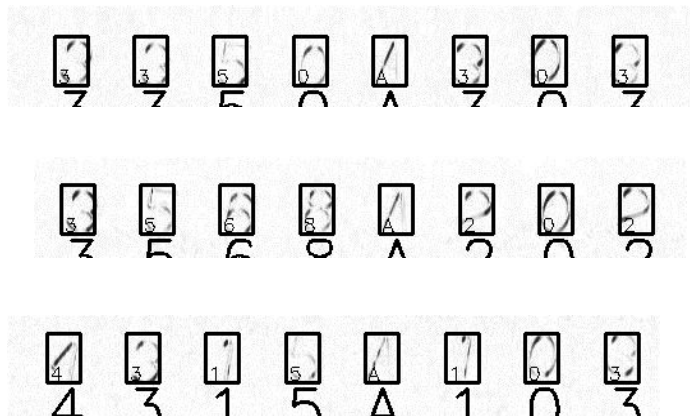


c) Generovaný hra-
nový obraz



d) Hrany (zelené) a
delta-fáza (modrá)

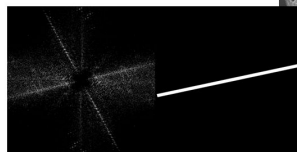
Industrial OCR



Algorithm - 2.Step

2. Angle rotation
(detection of the main direction)

www.joanneum.at



Input image

FFT spectrum

Detected main direction

Pedestrian detection, tracking in video

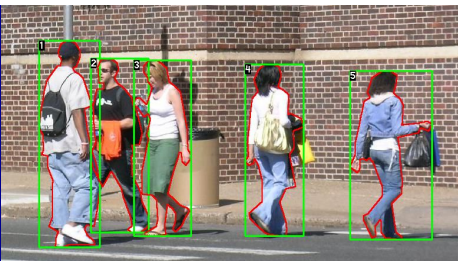


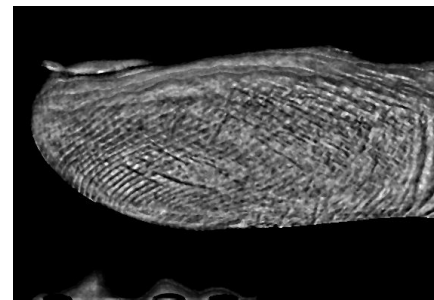
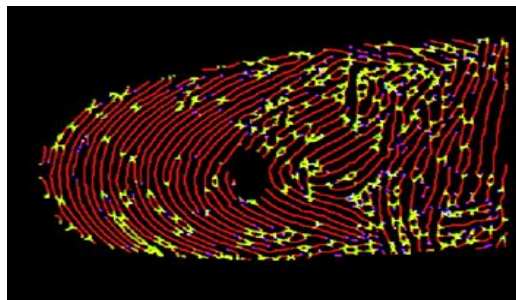
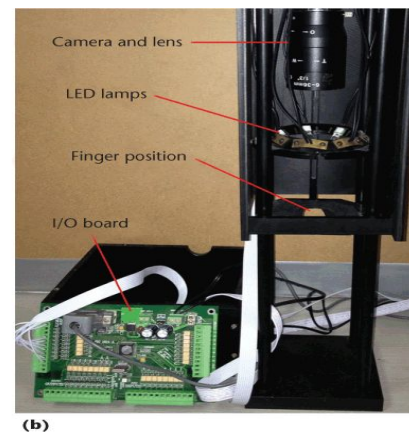
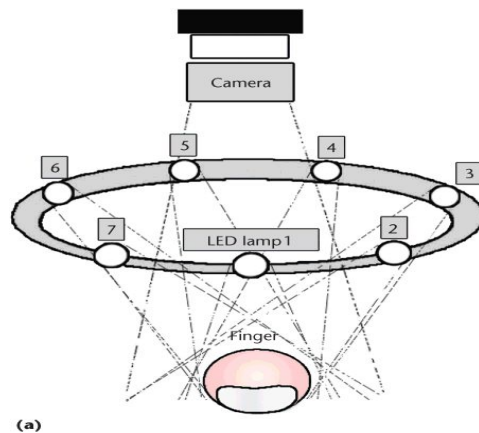
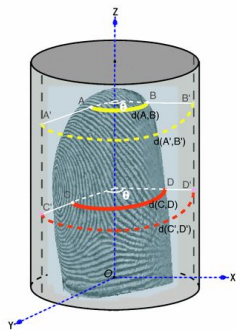
Figure 1. Crowded Scene in an Underground Station. Pedestrians which cross the bright (blue) line at the foot of the elevator are being counted in this scenario.

Road sign detection



Security – project example

3D touchless fingerprint



Traffic applications

- Self-driving cars
- Car detection
- Pedestrian detection
- Traffic surveillance
- Traffic statistics
- Section speed measurement
- Accident detection

Self-driving cars: Problems

Where is the car?



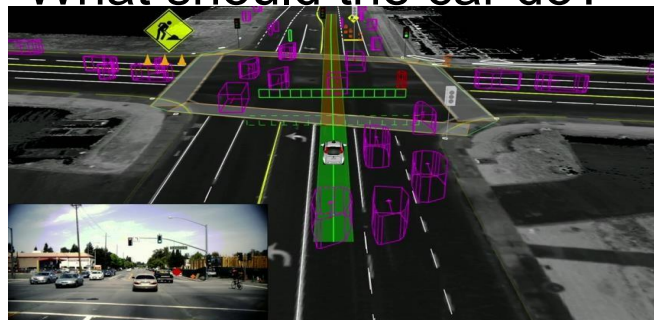
What is around the car?



What will happen



What should the car do?



Self-driving cars -> „deep driving“

Creating 3D model of the scene

Classifying and Detecting
Objects...

Deep Learning for Self-driving
Car



CV for Autonomous Driving

Sensor fusion, sensor registration

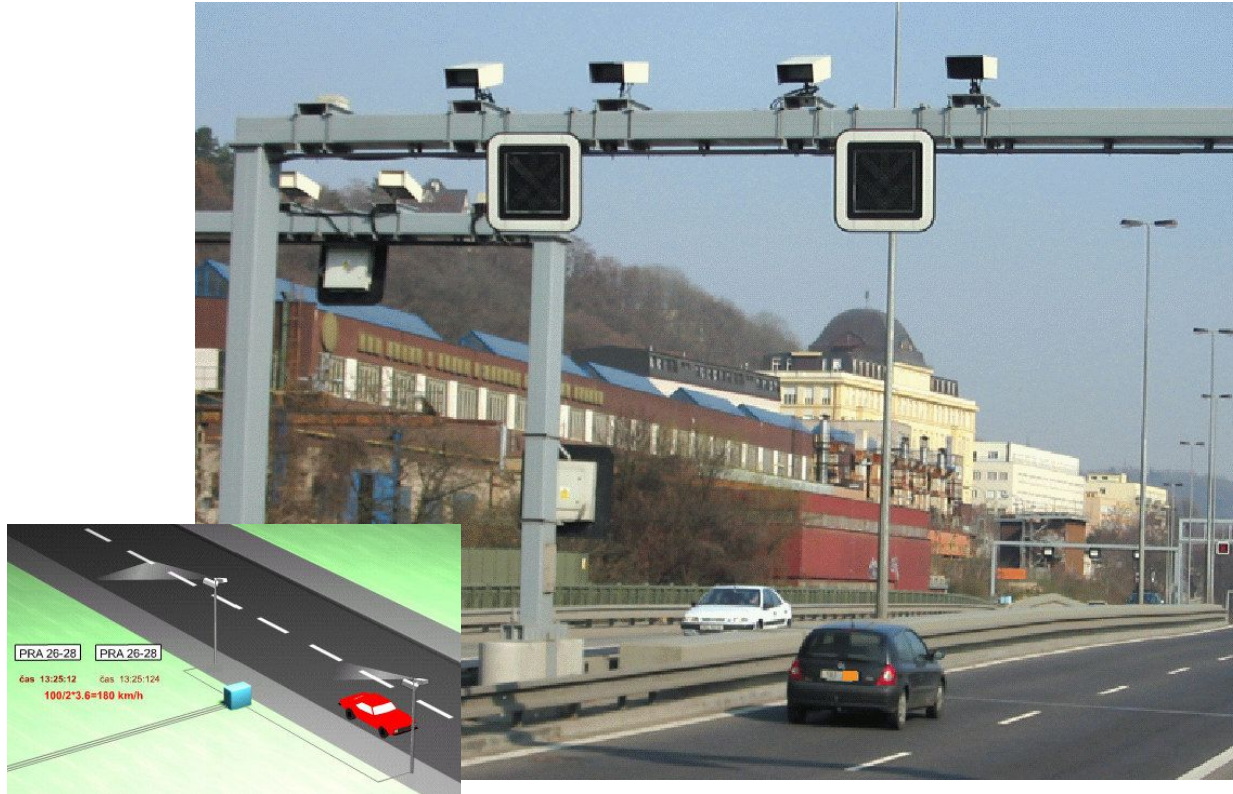
3D - Stereo vision, Localization, mapping

Semantic scene understanding

Deep Learning Methods

and more...

Section speed measurement



The Moon - NASA

The colors in this image are "enhanced," in the sense that the camera Galileo used to photograph the moon was **sensitive to near infrared wavelengths** of light beyond human vision.



Applications of computer vision in robotics

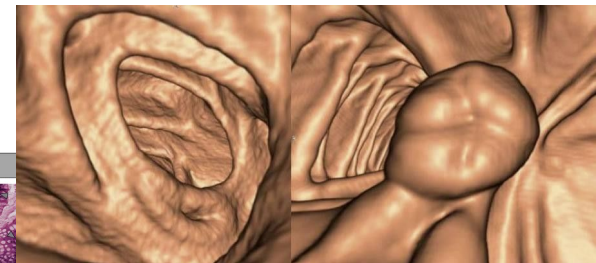
Example of industry application



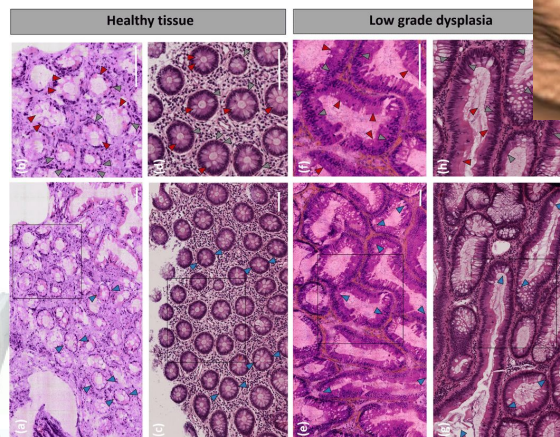
Medical imaging

Computer vision in medicine

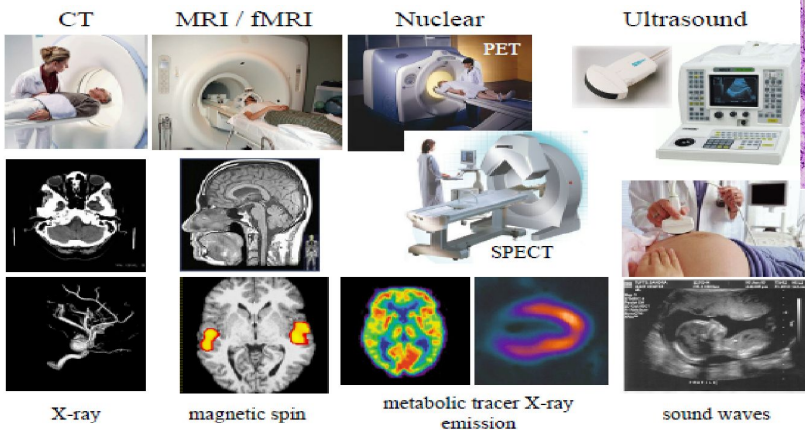
Visual inspection
Digital Pathology
Radiology



Virtual Colonoscopy



Histology

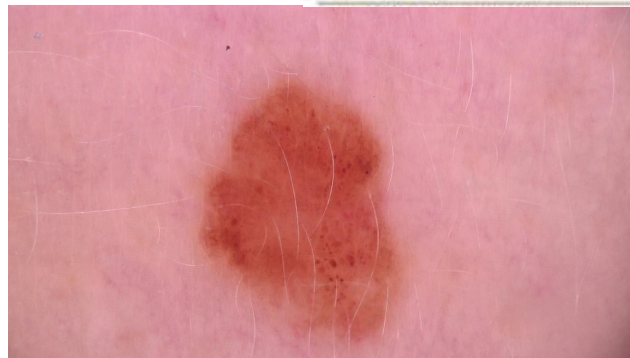
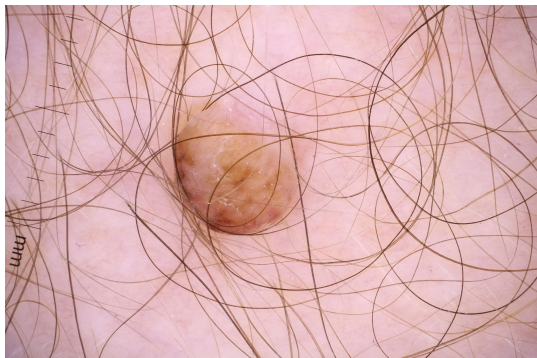
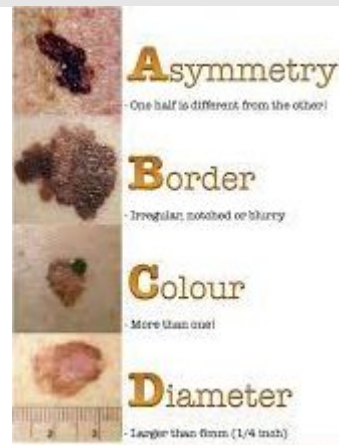


Radiology

Example of Visual Inspection: Dermoscopy

Identify melanoma in lesion images research

challenge: SIIM-ISIC Melanoma Classification



Example of Visual Inspection: Retinal Image Analysis

algorithms for the automated classification of different ocular diseases using retinal fundus images.

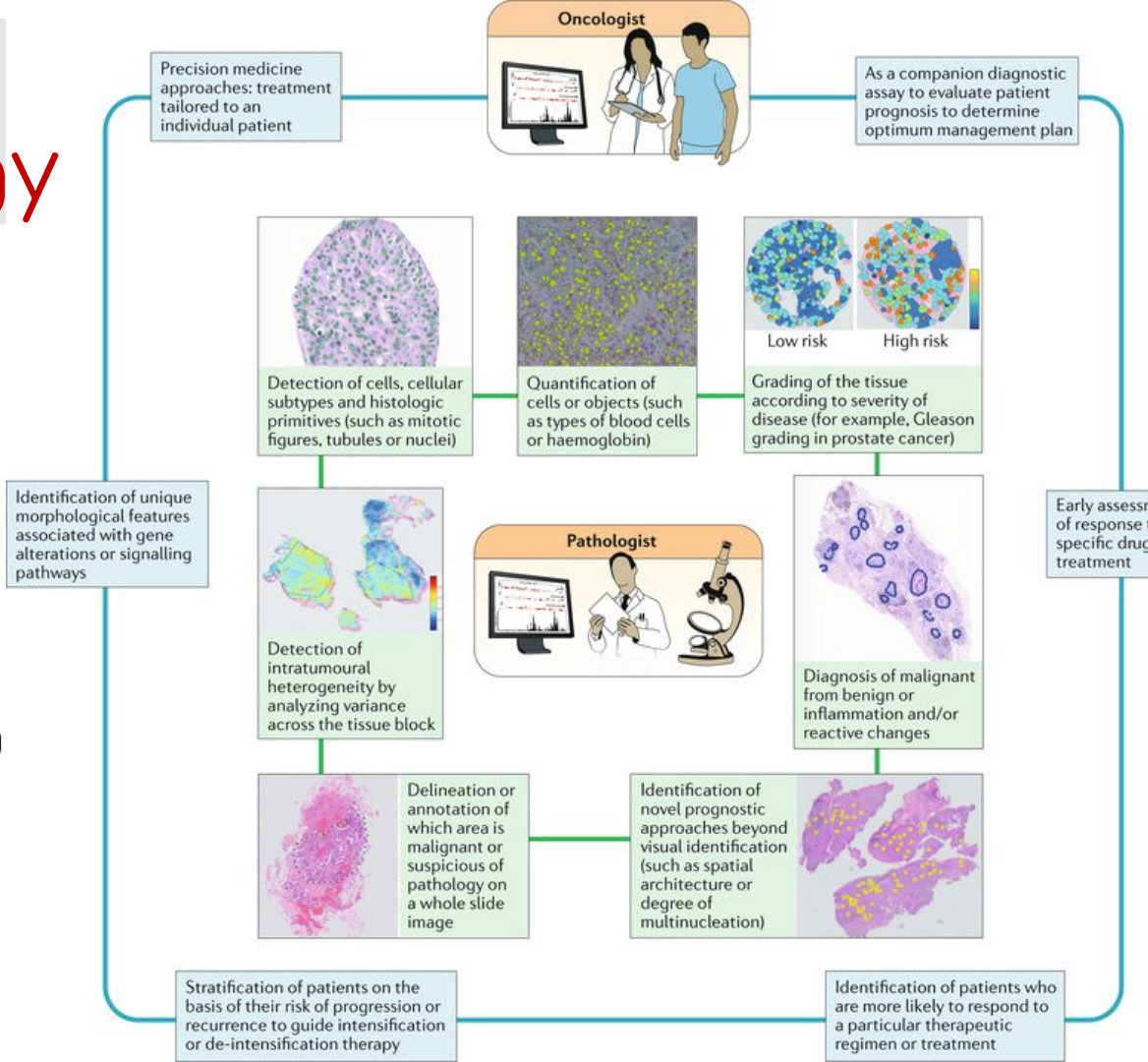


- primary focus on diabetic retinopathy, glaucoma
- age-related macular degeneration
- few other frequent pathologies

Example of digital pathology

Some of the AI approaches in digital pathology are shown.

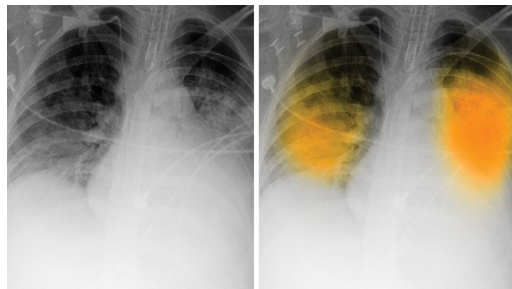
For practicing oncologists, AI can **assist** in decision-making across various aspects of cancer patient management.



Example of Radiology: Pneumonia in the lungs

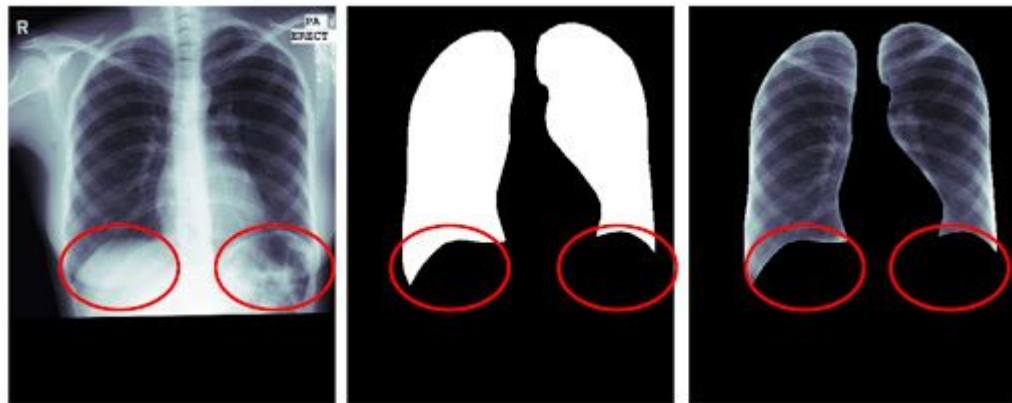
Scientists are developing a multitude of artificial intelligence algorithms to help radiologists, like this one that lights up likely pneumonia in the lungs.

ALBERT HSIAO AND BRIAN HURT/UC SAN DIEGO AIDA LABORATORY



RSNA Pneumonia Detection Challenge

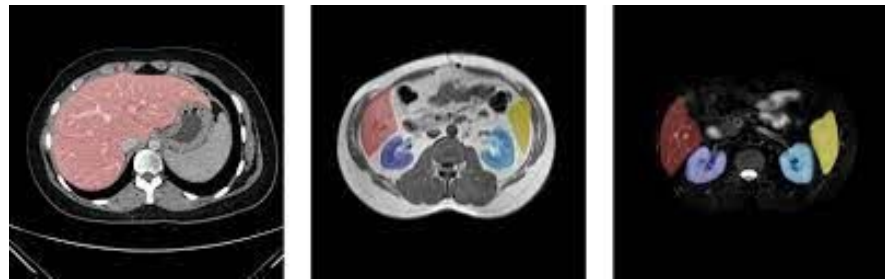
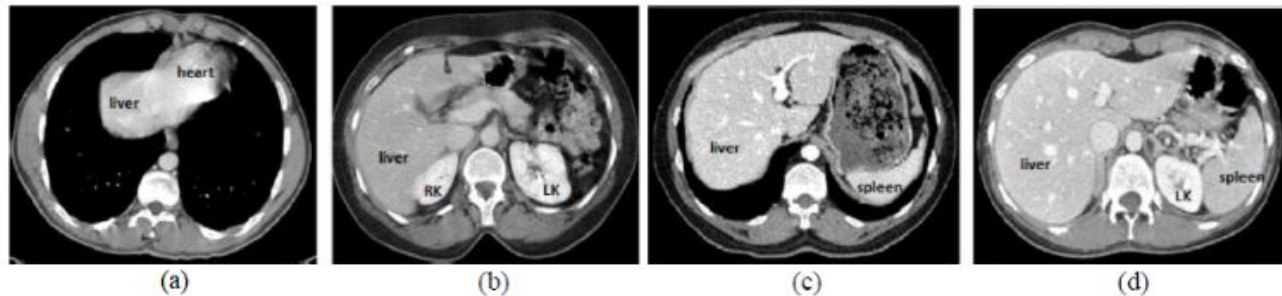
Can you build an algorithm that automatically detects potential pneumonia cases?



Pneumonia is an infection that inflames the air sacs in one or both lungs. The air sacs may fill with fluid or pus (purulent material), causing cough with phlegm or pus, fever, chills, and difficulty breathing. A variety of organisms, including bacteria, viruses and fungi, can cause **pneumonia**.

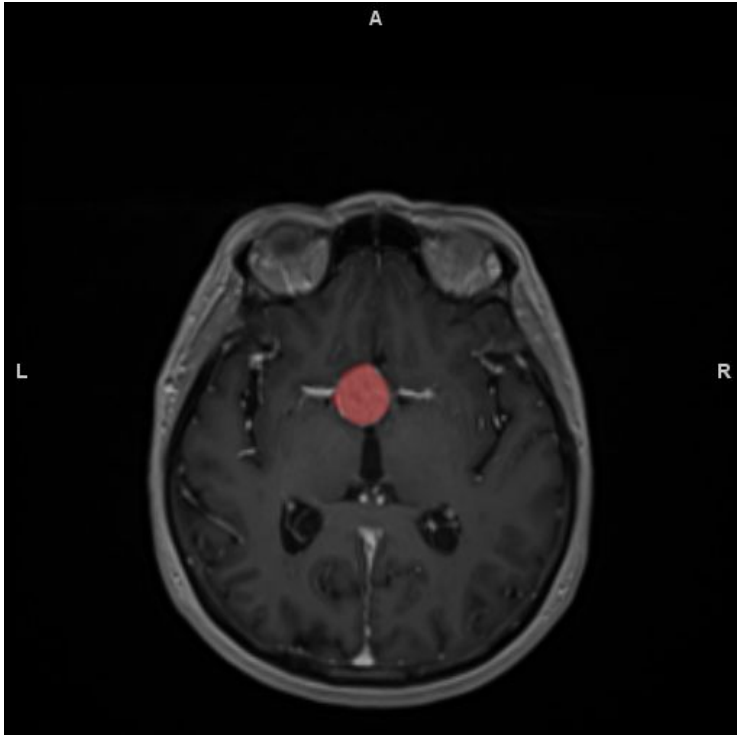
Combined Healthy Abdominal Organ Segmentation CT and MRI

CHAOS challenge aims the segmentation of abdominal organs (liver, kidneys and spleen) from CT and MRI data.



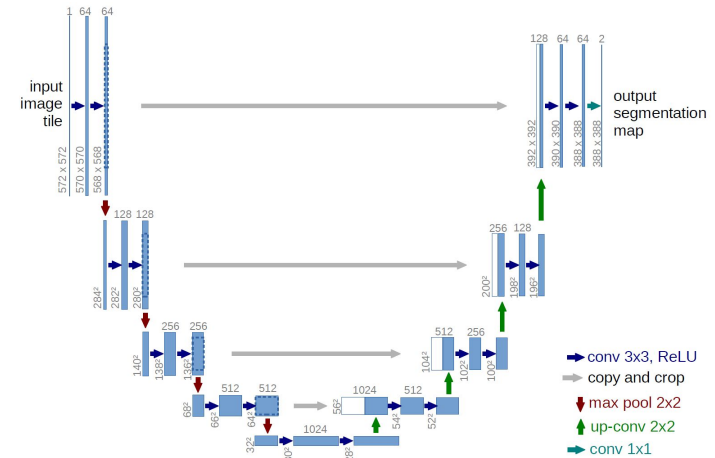
Example slices from CHAOS CT, MR (T1-DUAL in-phase) and MR (T2-SPUR) datasets (liver:red, right kidney:dark blue, left kidney:light blue and spleen:yellow).

Pituitary Adenoma Segmentation



NN: U-net

challenges:

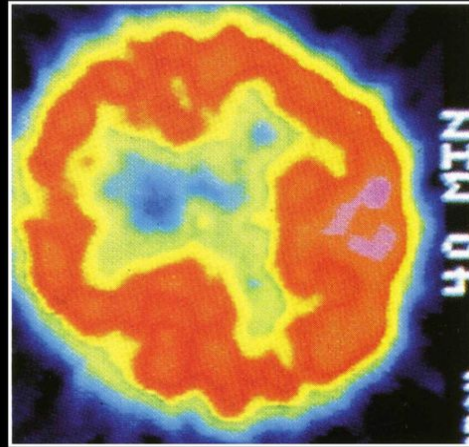


Positron emission tomography (PET)

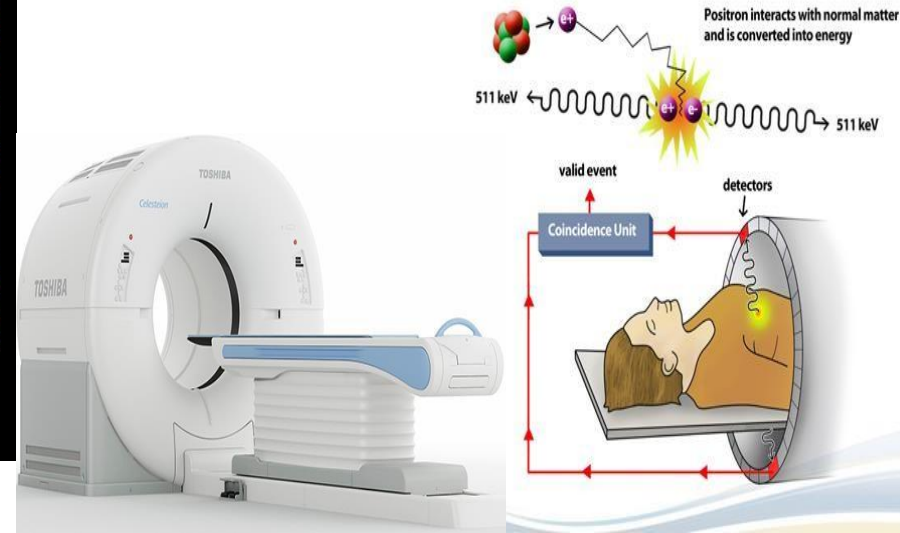
PET (*Positron Emission Tomography*)

SPECT (*Single Photon Emission Computed Tomography*)

- **Technique:** Positrons and photons are emissions from radioactive substances
- **What it shows:** An image of the amount and localization of any molecules that can be injected in radioactive form, such as neurotransmitters, drugs, tracers for blood flow or glucose use (which indicates specific changes in neuronal activity)



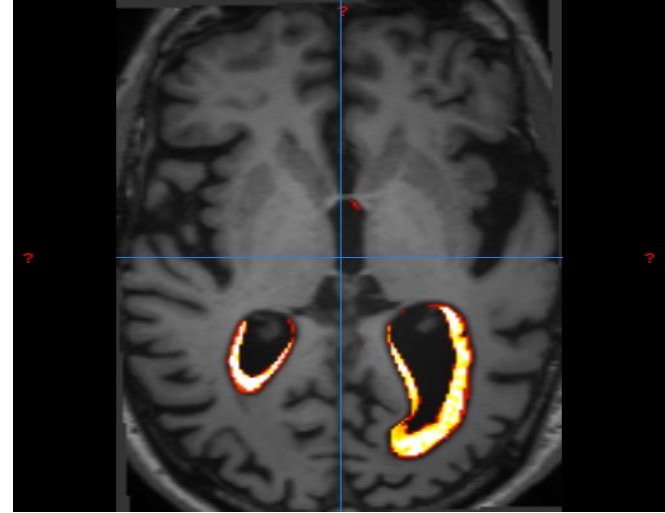
In PET a chemical compound labeled with a short-lived positron-emitting radionuclide of carbon, oxygen, nitrogen, or fluorine is injected into the body.



Alzheimer's disease

ADNI dataset

- Segmentation of anatomical parts of the brain using the U-net neural network
- Data: longitudinal studies - ADNI public dataset (MRI scans)
- Changes in the volume of anatomical segments are monitored over time:
 - lateral ventricles
 - hippocampus
 - amygdala

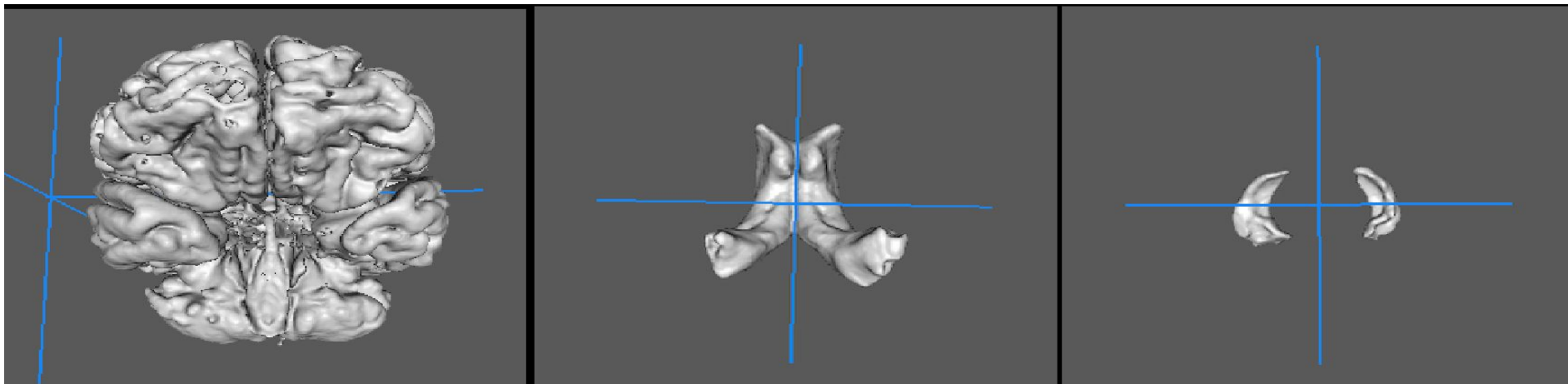


Longitudinal evaluation of data

Brain segmentation method based on deep neural networks

multistep brain segmentation approach based on 3D version of the U-Net neural network architecture.

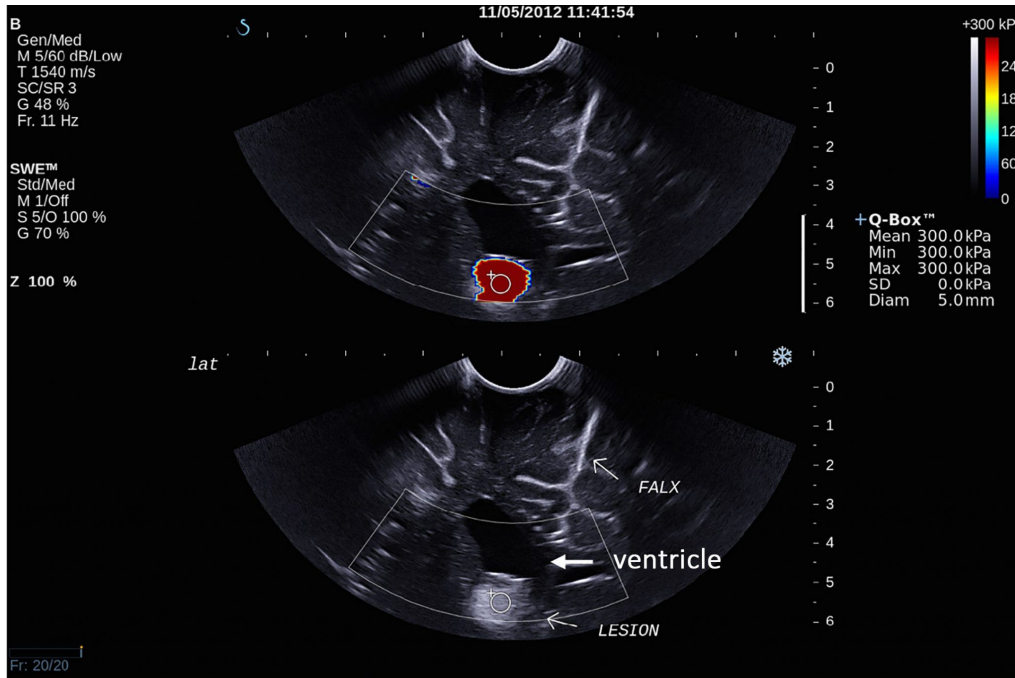
The network was trained using four different datasets (ADNI..) and tested on dataset OASIS.



Ultrasound

innovative type: shear wave elastography (SWE)

An innovative type of ultrasound scan that can detect cancer tissue left behind after a brain tumor is removed could improve the outcome from operations and reduce the risk of relapse, according to a new study published in the journal *Frontiers in Oncology*.



The technique, called shear wave elastography (SWE),

Intra-operative SWE for patient showing the tumor lying under the ventricles.

CV and Medical imaging – why?

- Save time
 - E.g. manual segmentation takes huge amount of time
- Prevent human failure
 - E.g. second opinion
- Recommend something
 - E.g. “look there, there’s some anomaly”
- Simplify doctor’s life
 - Show him/her the important stuff

THE GOAL IS NOT TO REPLACE THE DOCTOR!

Historical milestones

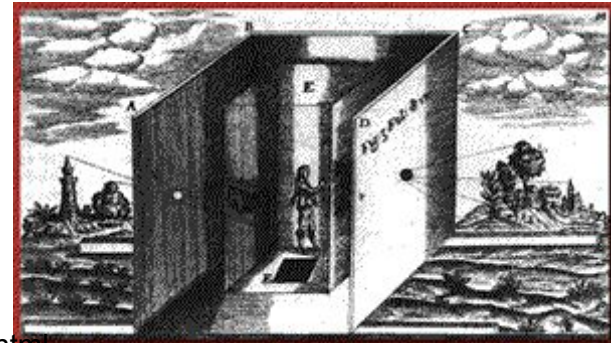
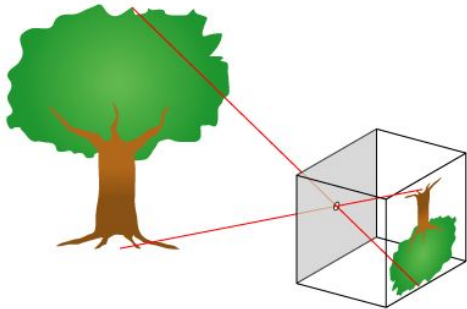
Historical milestones

Pinhole camera (Camera obscura)

Aristotle - in the 4th century BC...

Leonardo da Vinci - in 15th century

It was simply a darkened room, to which light was admitted only through a single small hole in the window shutter.



Historical milestones - Kinetoscope

1889 - THOMAS ALVA EDISON

Edison's Kinetoscope

The Kinetoscope was a device that gave the impression of movement by moving an endless loop of film continuously over a light source with a rapid shutter.

48 images/sec



Historical milestones – Digital Image Transmission

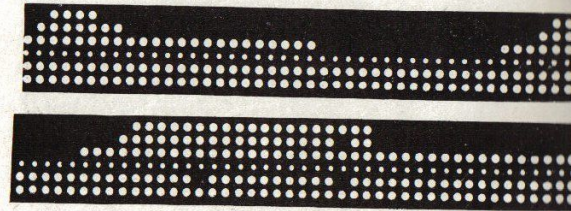
Bartlane cable picture
transmission system

Conversion in digital form –
5 Bit

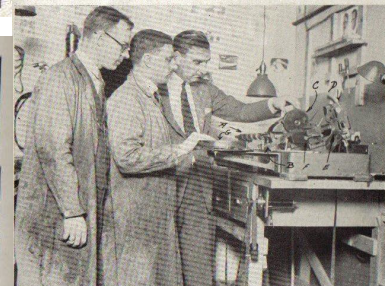
This system was invented
in 1920 and the first
transatlantic cable
picture was transmitted
in 1921, between London
and Halifax



Picture Transmitted and



Samples of the Baudot Tape Used to Transmit Pictures by
the Bartlane System.



The Bartlane Transmitter.

First Digital Image

1957

The photo is a digital scan of a shot initially taken on film.

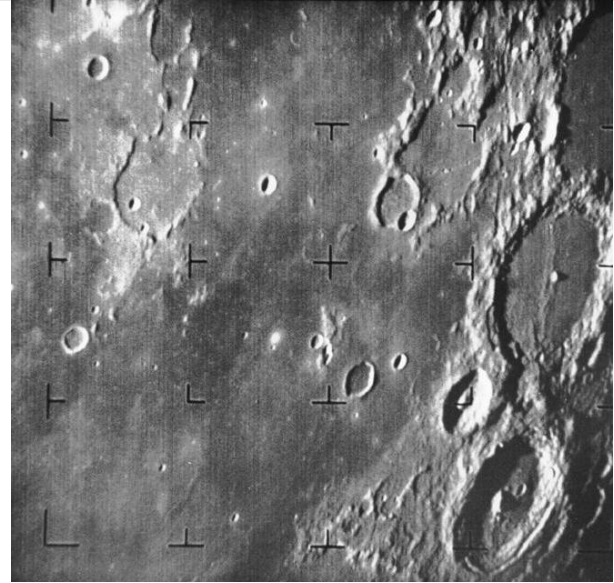


Historical milestones – Computer image processing

1964 : the birth of digital
image processing

Image processing using
computers

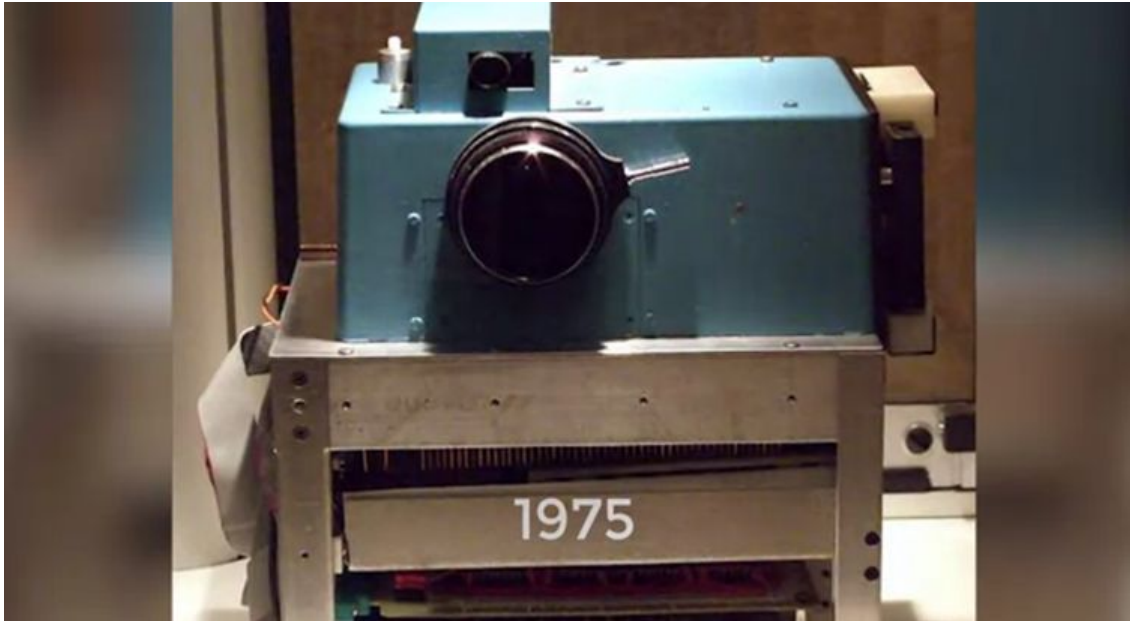
NASA project



The first picture of the moon by a U.S. spacecraft.
Ranger 7 took this image on July 31, 1964 (NASA.)

First digital camera

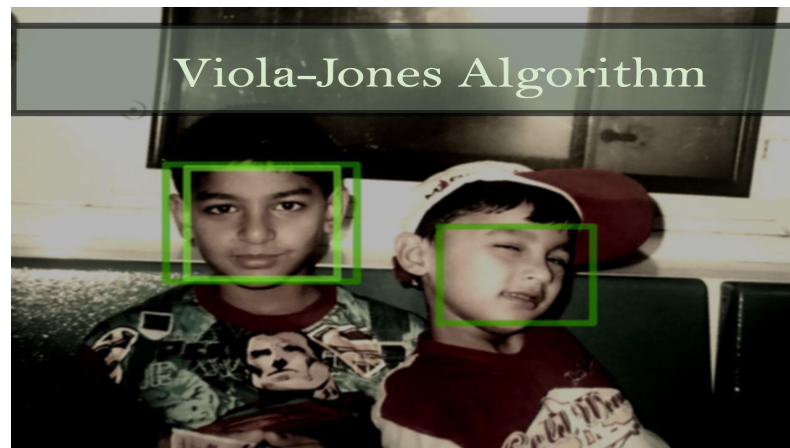
1975



Face detection i real time

2001

Paul Viola and Michael Jones

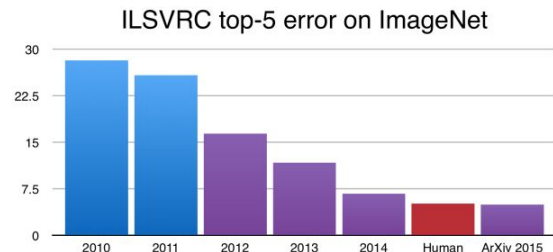
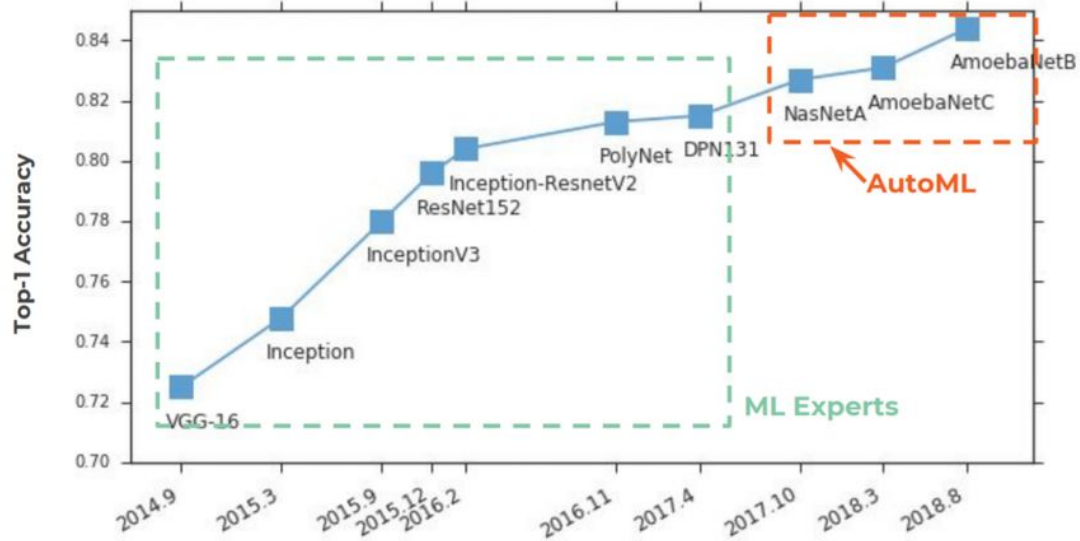


Viola, Paul, and Michael Jones. "Rapid object detection using a boosted cascade of simple features." Proceedings of the 2001 IEEE computer society conference on computer vision and pattern recognition. CVPR 2001. Vol. 1. IEEE, 2001.

Deep neural networks -> 2009

ImageNet database -> DNN

ImageNet



Visual data acquisition

What is an image?

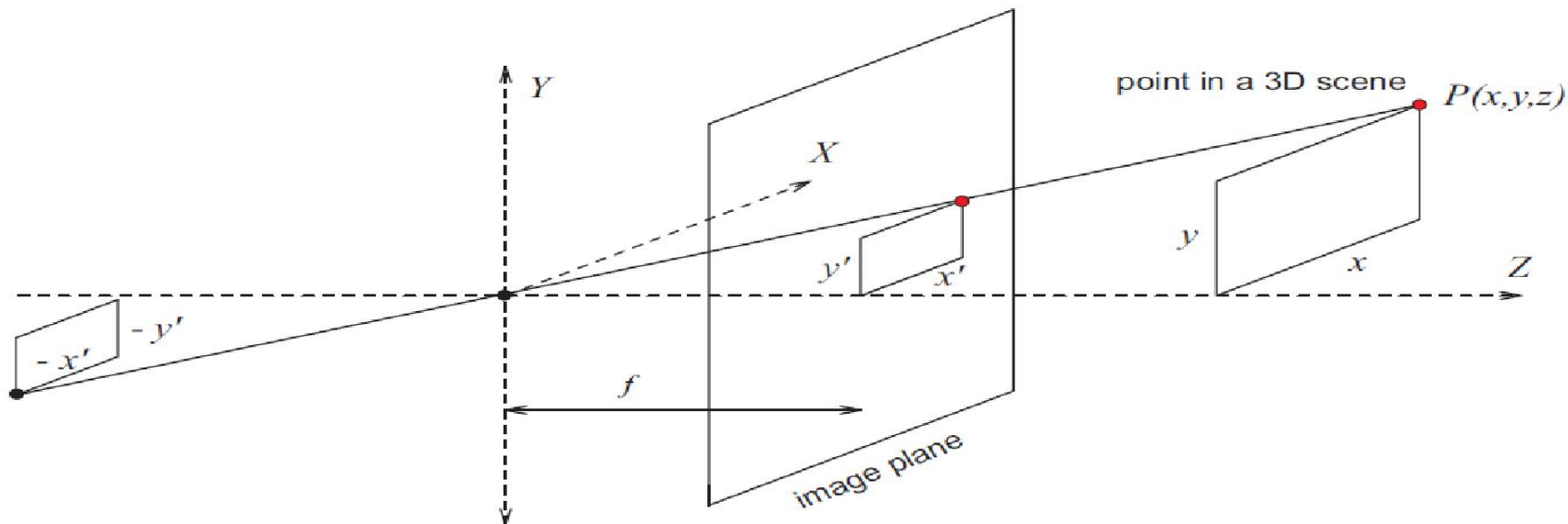
2-dimensional signal

A gray-scale image can be defined as a two-dimensional function $F=f(x, y)$

- x and y are spatial (plane) coordinates
- The amplitude F of $f(x, y)$ is the intensity of gray level
 - Intensity (jas, intenzita)
 - Pixel = the acronym from picture element.

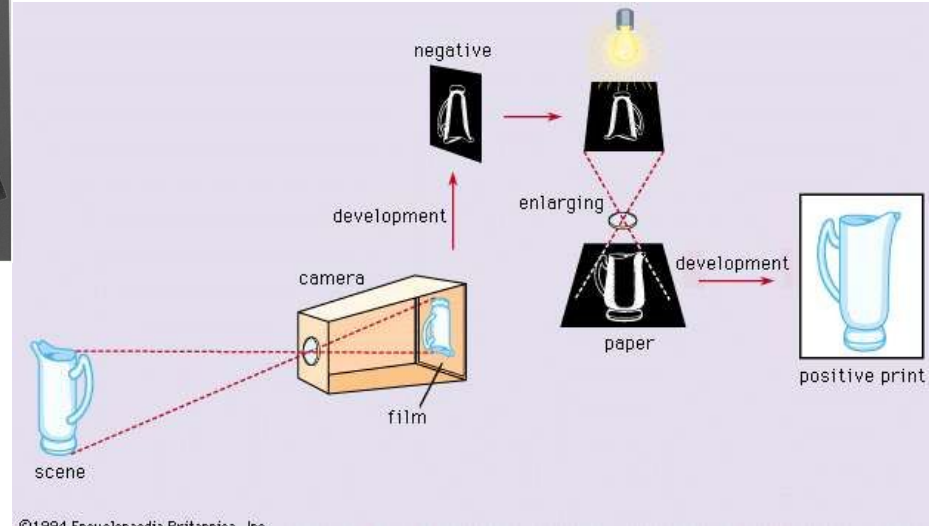
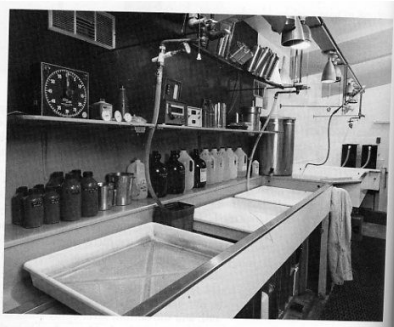
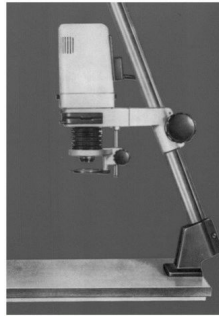
Image function $f(x, y)$,

Image function $f(x, y)$ or $f(x, y, t)$ is the result of the perspective projection.



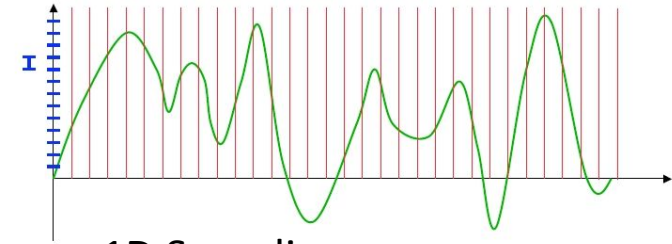
Analogue photography vs. digital photography

Technology of photography - analogue camera + film
light-sensitive material (negative, positive)

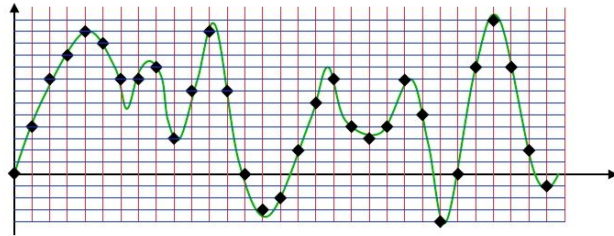


Analog to digital conversion

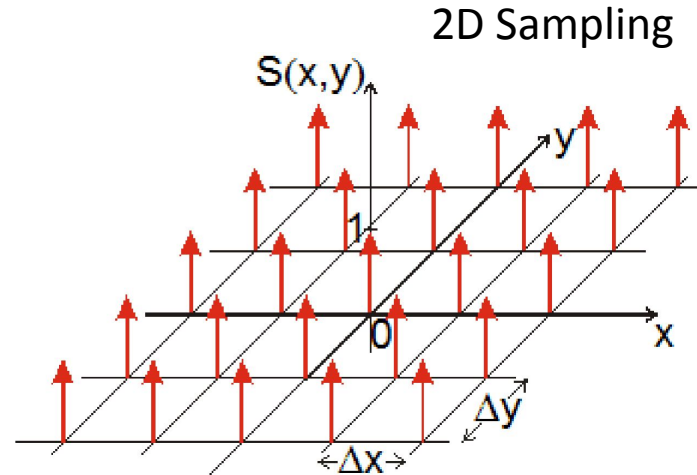
Digitalization: Sampling & Quantization



1D Sampling



1D Sampling & Quantization



2D Sampling

Dynamic Range & Gray-Level Resolution

Dynamic range describes the ratio between the maximum and minimum measurable light intensities (white and black, respectively)

The number of gray levels: L
after digitization is typically an integer power of 2

$$L = 2^k$$

Commercial Cameras typically: 8-12 bits

Dynamic range - Lena example



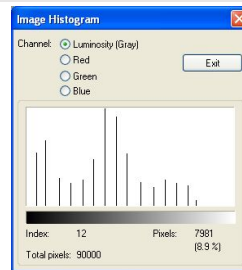
2
bits



3
bits



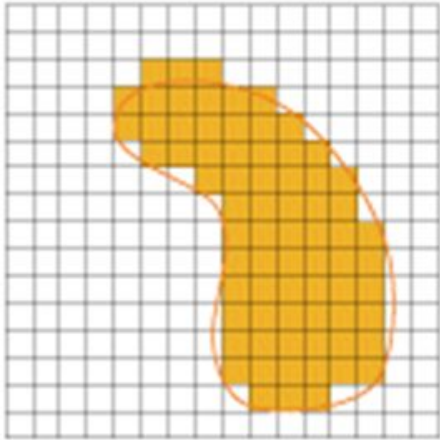
4
bits



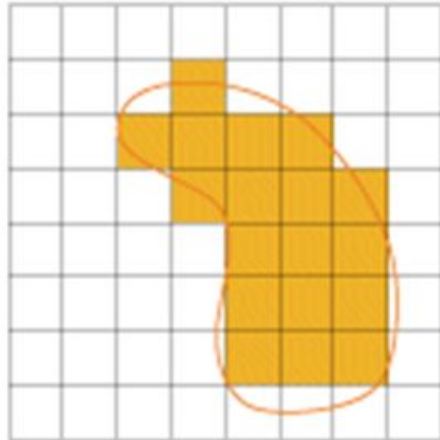
Spatial resolution

Number of cells in x and y direction

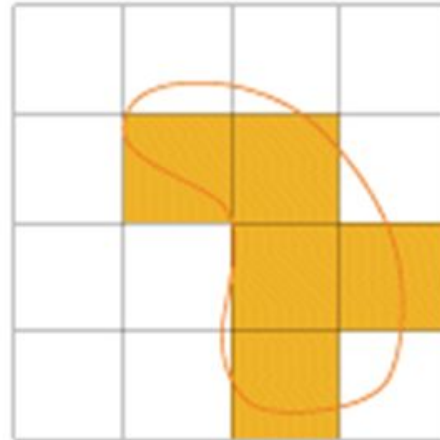
1 m cell
16 x 16 cells



2 m cell
8 x 8 cells



4 m cell
4 x 4 cells



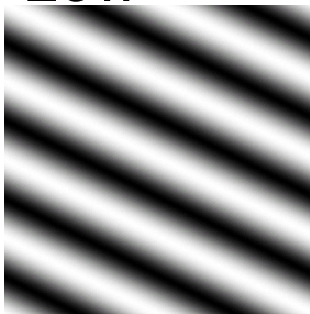
Nyquist sampling theorem

The sampling frequency should be at least twice the highest frequency contained in the signal.

- $f_s > 2 f_{\max}$

Frequency in the image – examples

Low ->



-> High



Sub-Nyquist sampled image – Moiré pattern

Sub Nyquist sampled image showing a Moiré pattern



Dimension of Image Matrix

Gray-scale image

$m \times n \times 1$

Colour image

$m \times n \times 3$

Colour image + alpha (opacity) channel

$m \times n \times 4$

Multi-spectral image

$m \times n \times x$

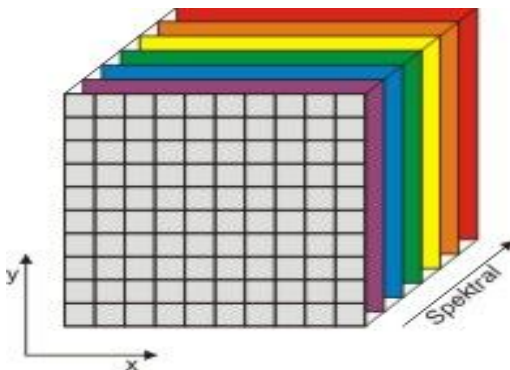


Image sequence – video

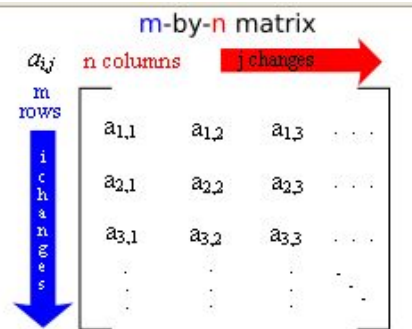
Additional dimension – time

Frequency of images [frames/sec]

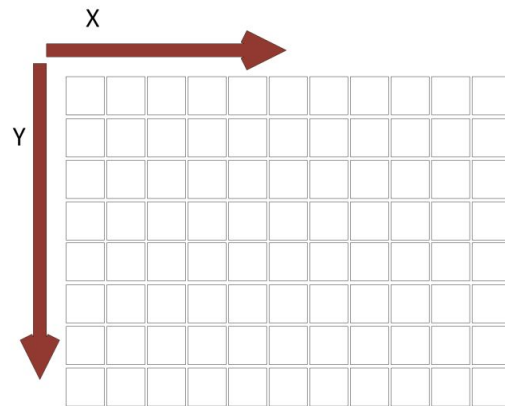
- Human sensing ~ approx. min. 10 frames/sec

Matrix Representing of Digital Image

The result of sampling and quantization is a matrix of numbers



=> *matrix operations*



Coordinate convention of an array (image libraries)

Source of Visual data

Visual data

Visible light

Infrared

Ultrasound

Radar

Röntgen, CT, MRI

Depth image

...and more

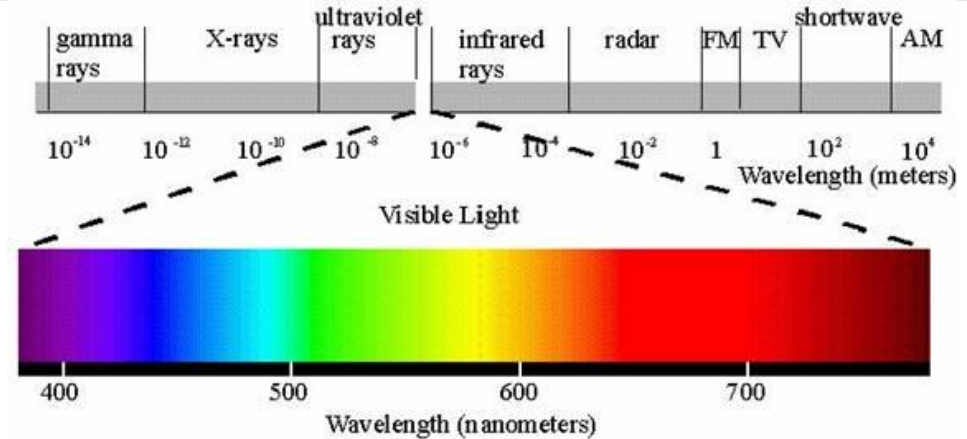
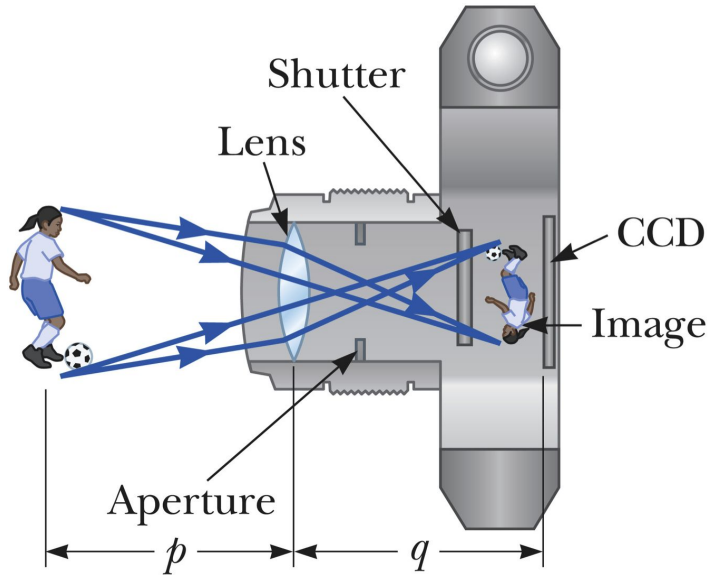


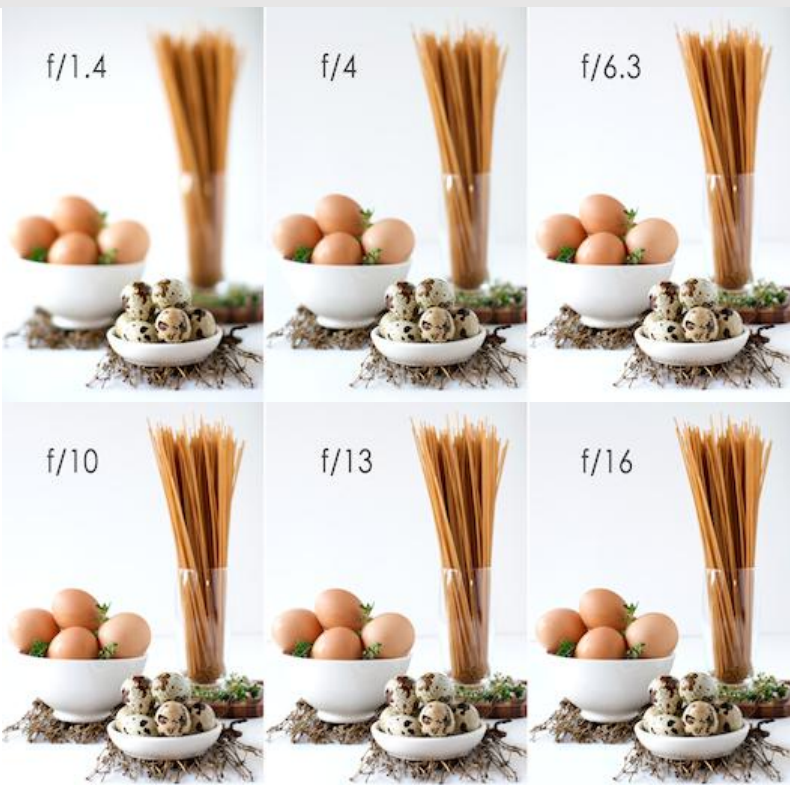
Image acquisition - digital camera



Optical sensor:

- CCD (Charged Coupled Device)
- CMOS (Complementary Metal Oxide Semiconductor)

Image acquisition - Depth of Field



Aperture

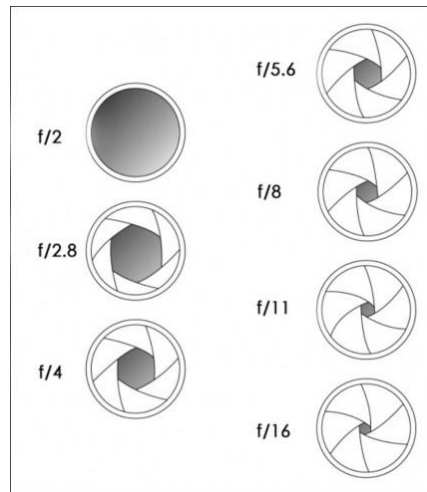
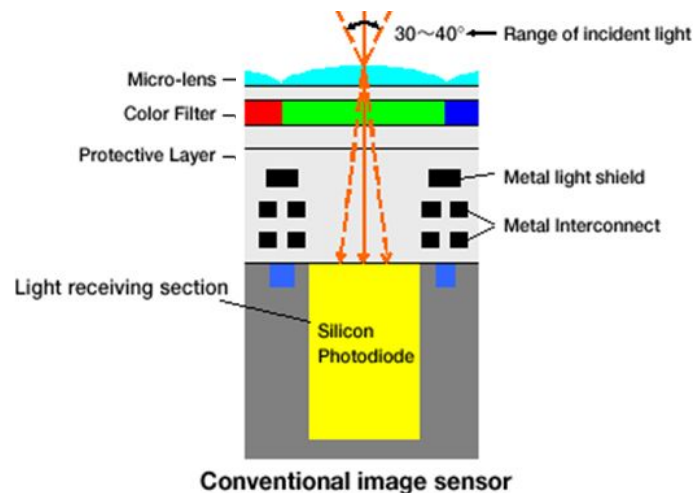
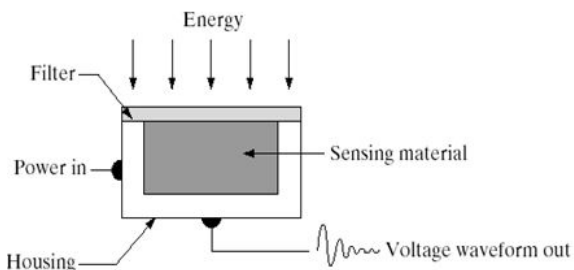


Image sensor – single unit

Image Sensor

- Physical device that is sensitive to the energy radiated by the object we wish to image

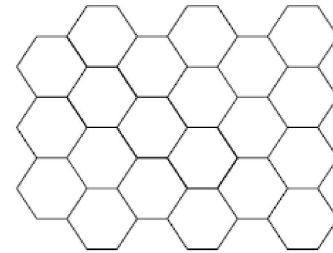
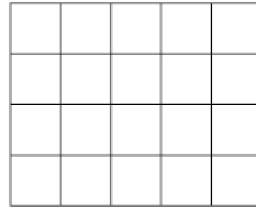
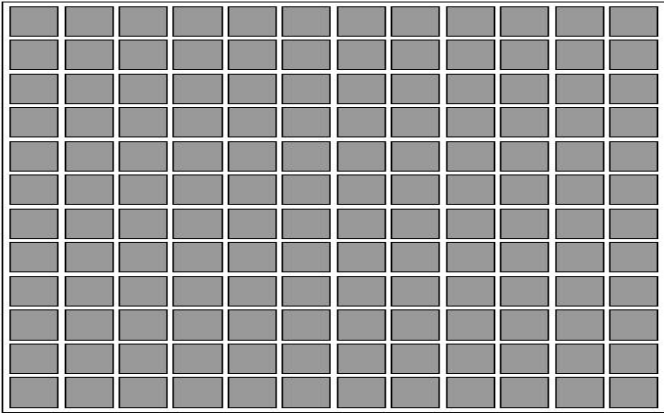


Principal Sensor Arrangements



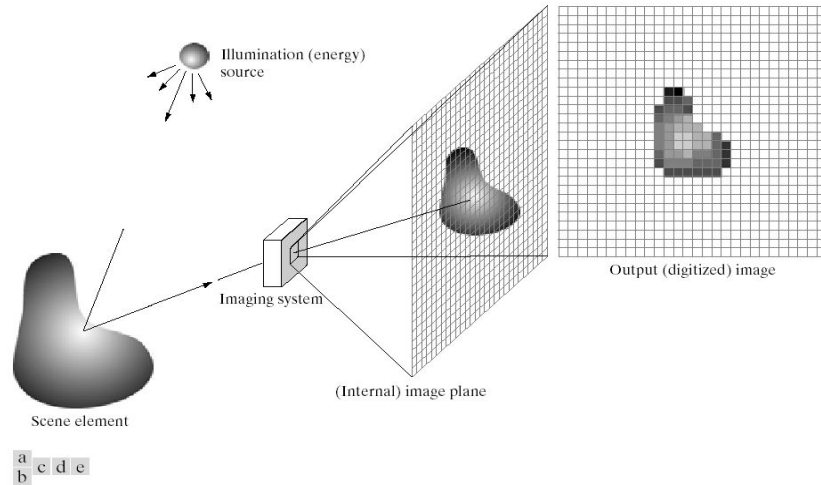
Line sensor

Array sensor



Arrangement of sampling points into a raster

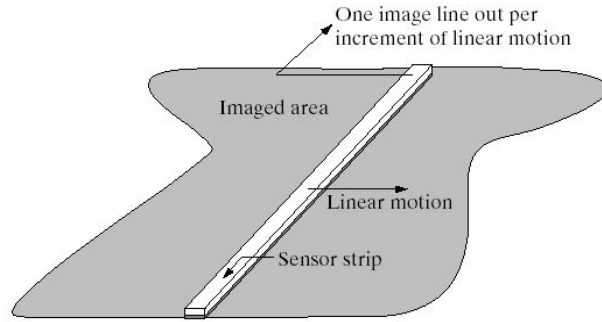
Image Acquisition Using Sensor Arrays



A typical sensor for the cameras is a CCD/CMOS array

- (a) Illumination source.
- (b) An element of a scene.
- (c) Imaging system.
- (d) Projection of the scene onto the image plane.
- (e) Digitized image.

Line scan camera

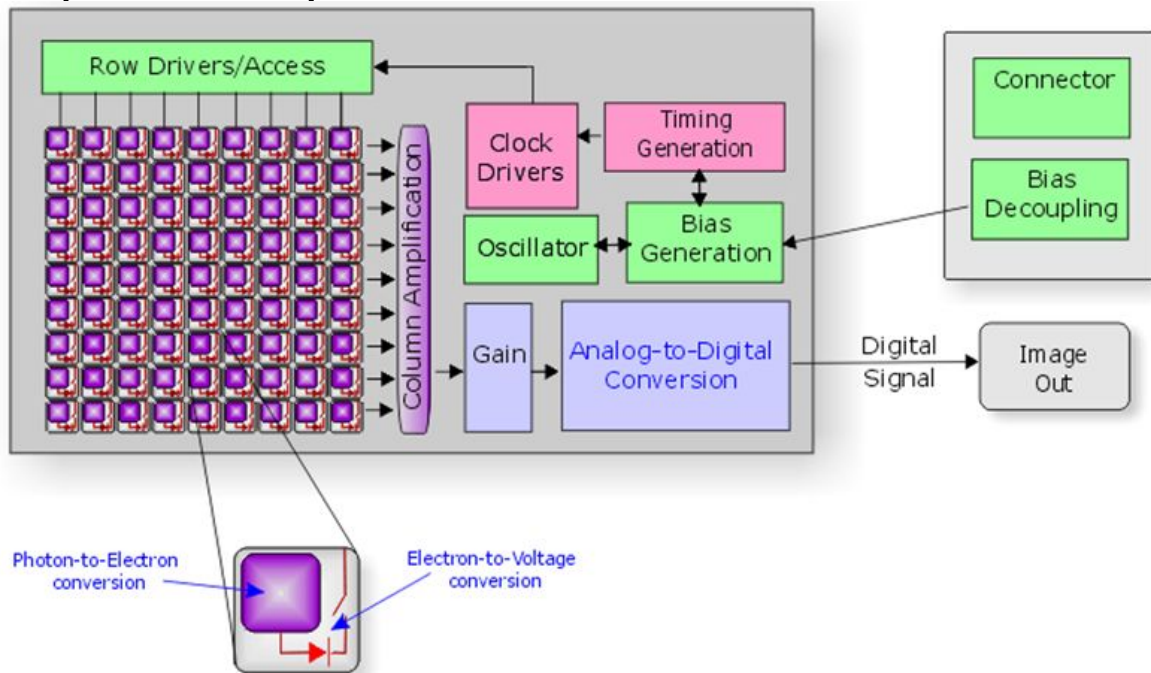


The imaging strip gives one line of an image at a time, and the motion of the strip completes the other dimension of a two-dimensional image.



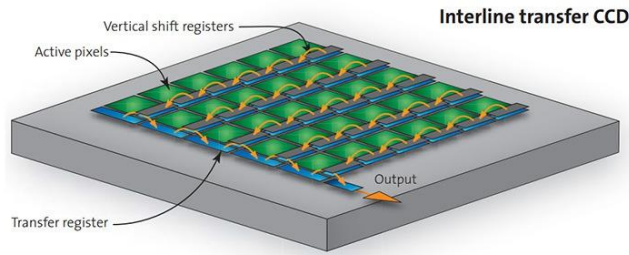
CMOS Sensor

Each pixel contains a photodetector – can be read separately



CCD- Charge-coupled device Image Sensor

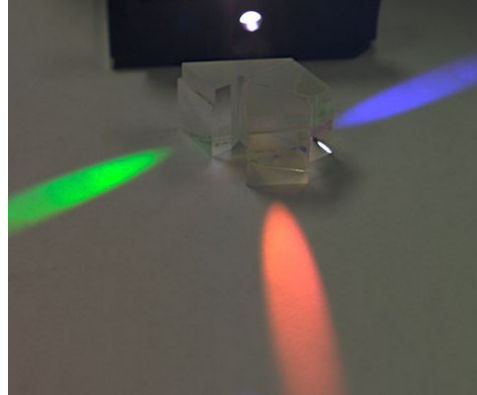
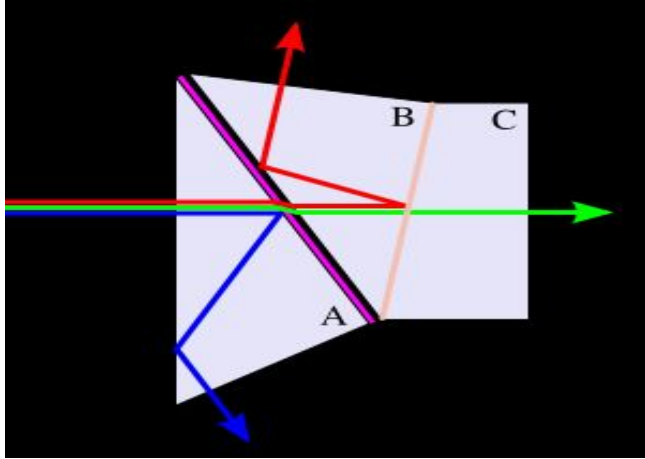
In a CCD device, the charge is actually transported across the chip and read at one corner of the array.



The first digital cameras used
CCD (Charged coupling Devices)

RGB -Prism Based Imaging

A color-separation beam splitter prism

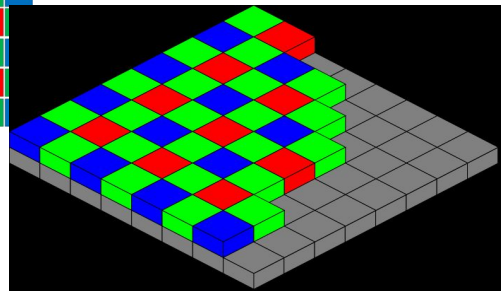
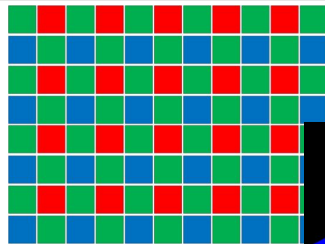


Three sensors
3CCD Colour line scan cameras

RGB -Bayer matrix

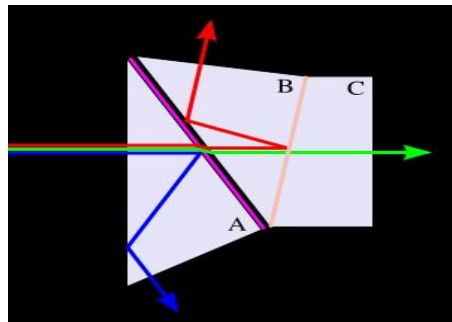
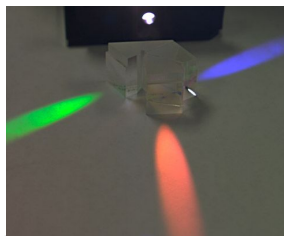
Bayer matrix

- The Bayer arrangement of color filters on the pixel array of an image sensor.



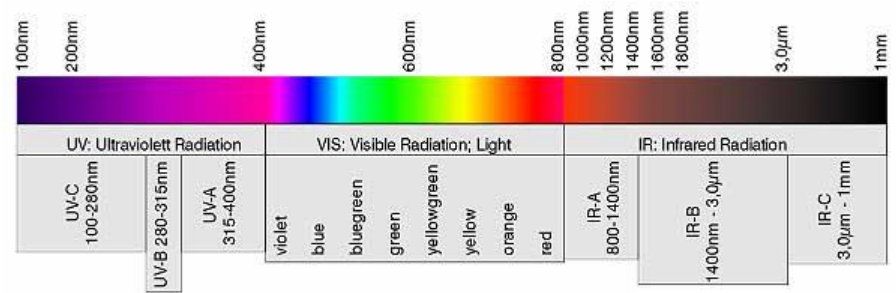
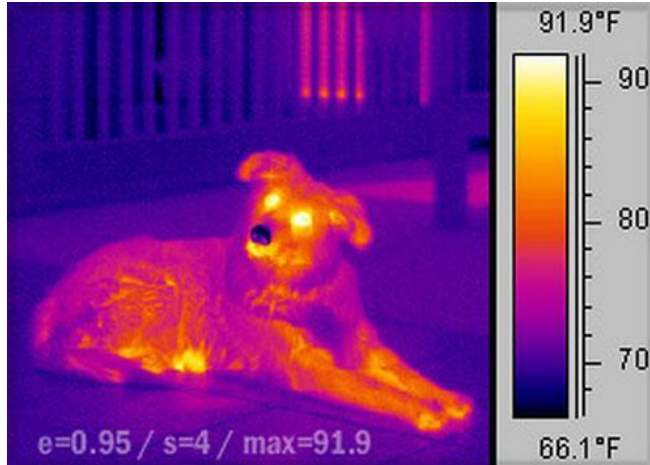
Three-CCD

- A color-separation beam splitter prism

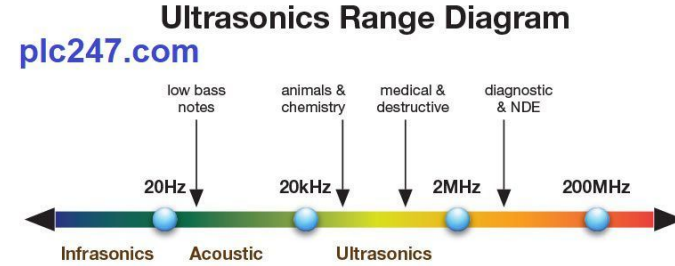
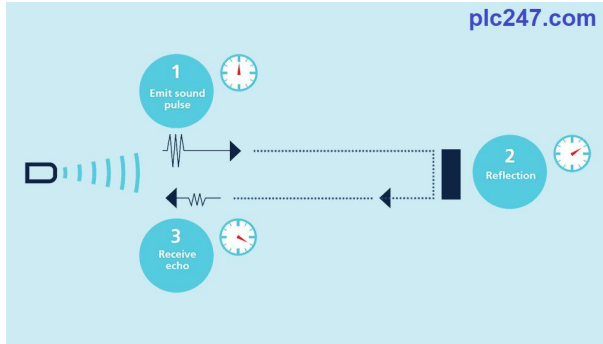


Infrared camera

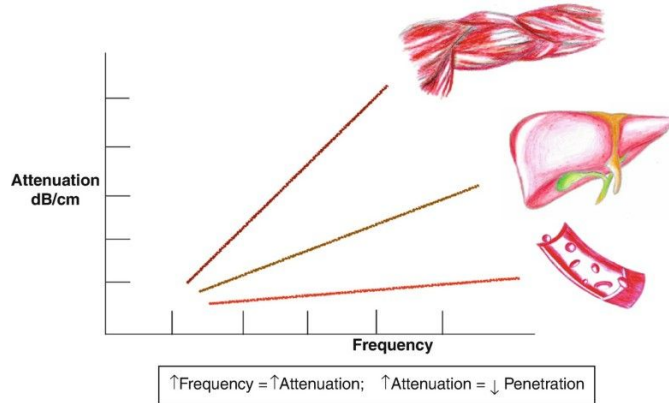
Infrared camera - a device that forms an image using infrared radiation



Ultrasonic Sensor



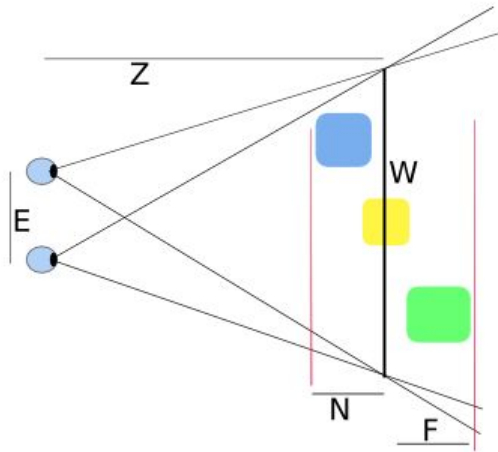
from Wikipedia article on Ultrasound



https://link.springer.com/chapter/10.1007/978-3-030-18371-4_1

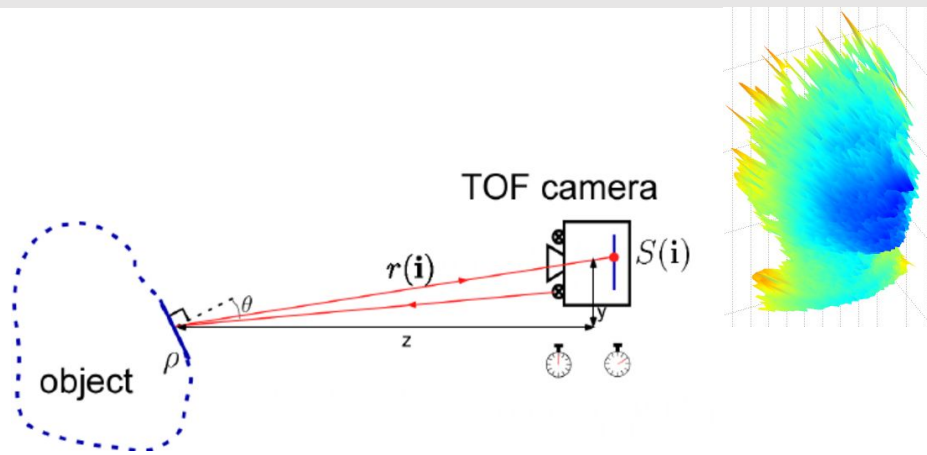
3D (2.5 D) image Acquisition Stereo camera

Stereo arrangement



TOF Time-of-flight (2.5 D camera)

- X MHz intensity-modulated light source
- Emitted light is reflected back to the camera by objects or persons within the scene
- Smart pixels in the sensor determine its **phase shift for each pixel**
- The phase shift is proportional to the distance of the captured object



IR pattern (2.5 D camera)

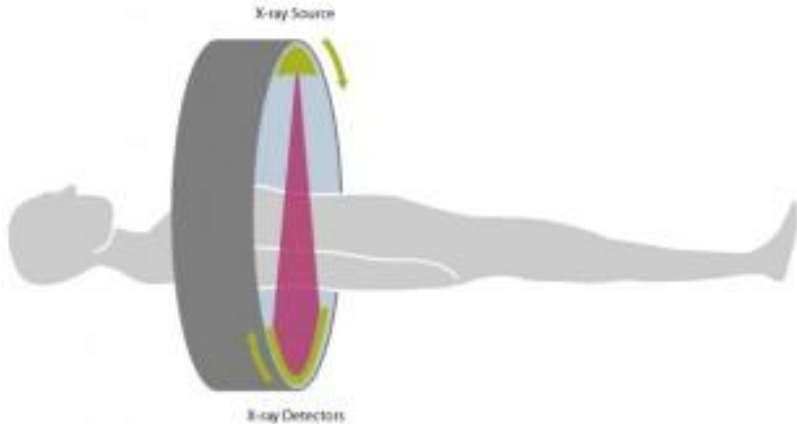
1. Emit IR pattern
2. Capture the IR image using CMOS camera with an IR-pass filter
3. Relative position of dots in the pattern – calculate depth displacement at each pixel position



Computed Tomography (CT)

The principle of computed tomography with an X-ray source and detector unit rotating synchronously around the patient.

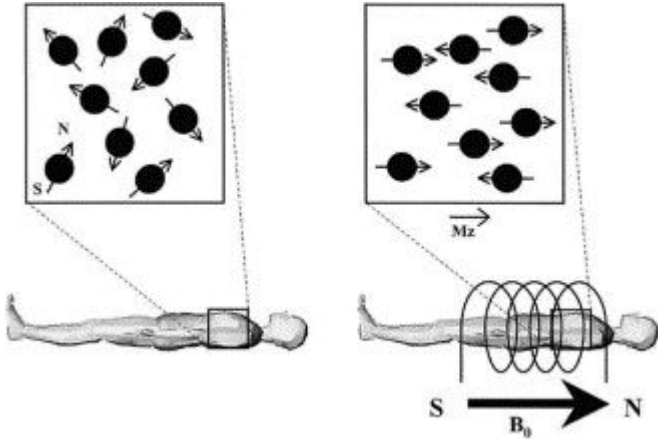
Data are acquired continuously during rotation.



Example of blood vessels and internal organs, by using relatively low radiation doses.

This CT scan of the whole thorax and abdomen was performed in less than 1 sec with only 20 mL of contrast media, and a radiation dose of 2.32 mSv.

Magnetic Resonance (MR)



Magnetic resonance imaging (MRI) is a noninvasive imaging technique

patient is submitted to a strong external magnetic field



OpenCV

https://docs.opencv.org/4.x/d9/df8/tutorial_root.html